

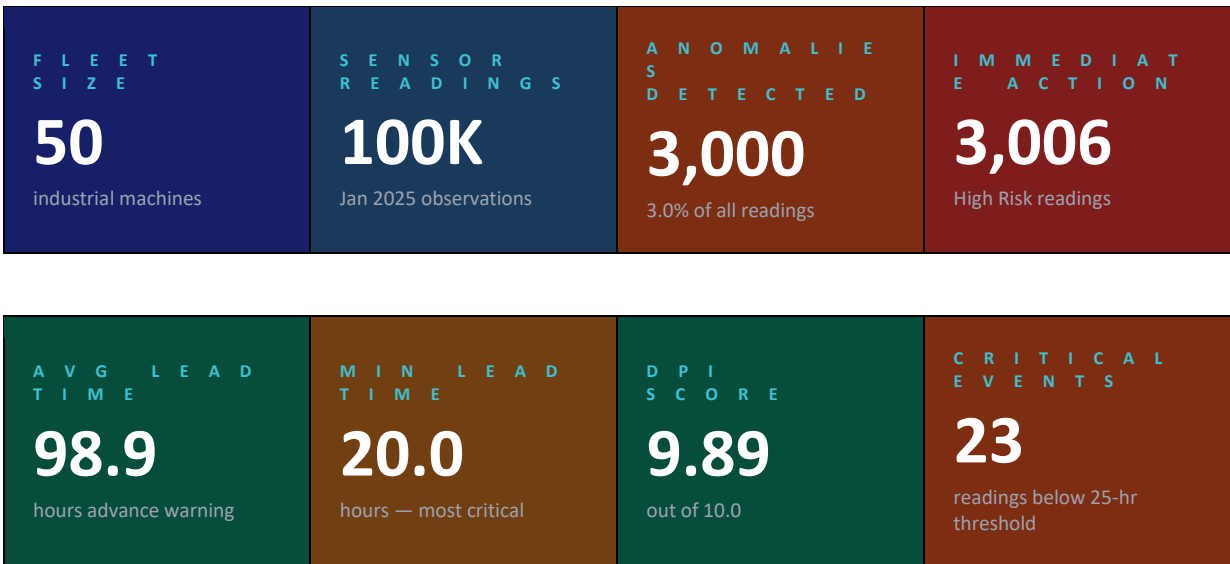
AI-Driven Predictive Maintenance & Anomaly Detection

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This executive summary presents findings from an AI-driven predictive maintenance study of the Thales Group manufacturing IoT fleet — 50 industrial machines, 100,000 sensor readings, January 2025. The study applies Isolation Forest and Autoencoder deep learning models to identify anomalous machine behaviour before failure, quantify intervention lead times, and detect cascade failure risk patterns across the production fleet.

At a Glance — Key Performance Indicators



1. Situation — What Was Found

The Thales Group manufacturing fleet of 50 machines generated 100,000 IoT sensor readings during January 2025. AI anomaly detection identified 3,000 confirmed fault events (3.0% of all readings) requiring immediate maintenance intervention. A further 30,998 readings (31.0%) are classified as Medium Risk requiring scheduled inspection within 72 hours.

Risk Level	Readings	Fleet %	Required Action
● High Risk — Immediate	3,006	3.0%	Immediate Maintenance Now
● Medium Risk — Schedule	30,998	31.0%	Inspect within 72 hours
● Low Risk — Monitor	65,996	66.0%	Standard Monitoring
TOTAL	100,000	100%	—

2. Critical Findings

Finding 1 — Machine M-25 Is the Highest Priority Target

Machine M-25 is anomalous on two independent dimensions simultaneously — it has both the lowest Anomaly Score in the entire fleet (0.0958) AND the highest Cascading Failure Index (1.48). This means M-25 is the most fault-prone machine AND the one most likely to trigger downstream chain failures across the production line. It must be treated as the highest-priority maintenance target fleet-wide.

Finding 2 — 98.9-Hour Average Lead Time Enables Planned Intervention

The predictive model delivers an average of 98.9 hours advance warning before predicted machine failure — nearly four full working days. This is sufficient time to source spare parts (typical delivery 24–48 hours), schedule specialist maintenance personnel, and coordinate with production planning to minimise output disruption. Traditional reactive maintenance provides zero lead time; this system transforms unplanned stoppages into planned interventions.

Finding 3 — 5,796 Emerging Cascade Readings Demand Immediate Scheduling

Temporal risk escalation analysis identified 273 Active Cascade Risk readings (87.2% anomaly rate — nearly all confirmed faults) and 5,796 Emerging Cascade readings. Emerging Cascade machines are exhibiting 2 of 3 warning conditions before reaching Active status. Intervening at the Emerging Cascade stage prevents both the machine failure AND the downstream production cascade. This is the most cost-effective intervention window available.

Finding 4 — Power Consumption Is the Primary Early-Warning Signal

Correlation analysis across all 50 machines identifies Power_Consumption_kW as the strongest predictor of anomalous machine behaviour (Pearson $r = 0.347$). Abnormal electrical load patterns precede visible mechanical symptoms, making power deviation monitoring the most reliable early-warning indicator. This has significant practical value — power monitoring is standard infrastructure in all smart metering-equipped facilities.

3. Top Risk Machines — Priority Matrix

Machine	Anomaly Score	HR Readings	Cascade Index	Risk Classification	Action
M-25	0.0958	77	1.48	 Highest Systemic Risk	NOW
M-29	0.0961	68	1.44	 High Risk	NOW
M-4	0.0961	74	1.45	 High Risk	NOW
M-14	0.0962	67	1.43	 High Risk	24 hrs
M-42	0.0963	68	1.46	 High Risk	24 hrs

Note: Lower Anomaly Score = more anomalous in Isolation Forest convention. M-25 carries both the lowest score (most anomalous) and highest Cascade Index (greatest propagation risk) — the only machine classified as maximum risk on both dimensions.

4. Cascade Failure Risk — Temporal Analysis

A three-condition temporal analysis detects developing chain failure patterns before they manifest as production-stopping events. Each condition captures a different dimension of deteriorating machine health:

- ▶ **Condition 1 — Repeated Faults:** Machine has ≥ 2 anomaly events in its most recent 7 sensor readings (persistent fault, not random spike)
- ▶ **Condition 2 — Worsening Trend:** Rolling Anomaly Index rising AND Risk Escalation Rate > 0.05 (confirmed deterioration trajectory)
- ▶ **Condition 3 — Propagation Risk:** Cascading Failure Index in top 10% of fleet (fault will spread to downstream machines)

Pattern	Count	Fleet %	Anomaly Rate	Urgency
● Active Cascade Risk (all 3 conditions)	273	0.3%	87.2%	Immediate
○ Emerging Cascade (any 2 of 3)	5,796	5.8%	28.4%	48 hours
● No Pattern (0–1 conditions)	93,931	93.9%	1.2%	Monitor

The 87.2% anomaly rate in Active Cascade versus 1.2% in No Pattern validates the detection logic. The 5,796 Emerging Cascade readings are the most valuable intervention target — these machines are showing clear warning signs before reaching critical status.

5. AI Models Used

Model	Approach	Key Output
Isolation Forest	Unsupervised tree ensemble — isolates anomalies by their property of being statistically rare	Anomaly Score (–ve = anomaly) Normalized Score 0–100
Autoencoder (Deep Learning)	Trained on normal data only — detects anomalies as high reconstruction error patterns	AE Anomaly Score 0–100 Reconstruction Error (MSE)
Ensemble Model	Average of both model scores — readings flagged by both are highest confidence	Ensemble Score 0–100 Reduces false alarm rate

6. Recommended Action — Three-Tier Maintenance Protocol

TIER 1 — IMMEDIATE (0–24 hours)

3,006 High Risk readings across the fleet | Priority: M-25, M-29, M-4 | 273 Active Cascade Risk events

- ▶ Deploy maintenance teams to M-25 immediately — highest combined anomaly + cascade risk
- ▶ Inspect M-29 and M-4 within the same shift — next two highest priority machines
- ▶ For all 273 Active Cascade Risk events: halt machine if possible to prevent production line cascade

TIER 2 — SCHEDULED (24–72 hours)

5,796 Emerging Cascade readings | 30,998 Medium Risk readings | Schedule inspection before pattern escalates

- ▶ Schedule all Emerging Cascade machines for inspection before 48-hour deadline
- ▶ Order spare parts now — use the 98.9-hour average lead time advantage
- ▶ Coordinate with production planning to minimise output disruption during scheduled windows

TIER 3 — CONTINUOUS MONITORING (72+ hours)

93,931 No Pattern readings | 65,996 Low Risk readings | Automated alert escalation if conditions develop

- ▶ Continue automated sensor monitoring — no immediate human intervention required
- ▶ Configure alerts for any transition from No Pattern to Emerging Cascade status
- ▶ Monitor Power_Consumption_kW deviation as primary early-warning signal across all machines

7. Policy Recommendations for Government Stakeholders

- ▶ **6G Manufacturing Infrastructure:** Mandate 6G-readiness in new industrial facilities — the 98.9-hour lead time demonstrated here requires continuous low-latency IoT connectivity
- ▶ **National PdM Data Standard:** Establish interoperability standards for industrial IoT sensor data to enable cross-sector predictive maintenance benchmarking
- ▶ **BEE Energy Audit Extension:** Power_Consumption_kW is the primary anomaly predictor ($r=0.347$) — extend Bureau of Energy Efficiency audit frameworks to include power deviation monitoring as an energy efficiency and fault prevention tool
- ▶ **Cascade Failure Regulation:** Develop regulatory standards requiring Cascading Failure Index disclosure for critical infrastructure manufacturing facilities
- ▶ **Workforce Development:** Fund Industrial Data Science programmes combining domain engineering knowledge (Electrical, Mechanical) with machine learning — this study demonstrates the value of that hybrid skill profile

8. Strategic Conclusion

The Thales Group AI predictive maintenance system delivers a Downtime Prevention Index of 9.89/10 — near-maximum performance — with an average 98.9-hour early warning window that transforms reactive maintenance into planned intervention. The system identified 3,000 fault events, 273 cascade chain-failure risks, and flagged Machine M-25 as the single highest systemic risk in the fleet.

For government stakeholders: this study provides robust evidence that AI-based predictive maintenance, when integrated with 6G IoT infrastructure, constitutes a critical industrial competitiveness tool. The three-tier maintenance protocol recommended here is directly implementable with existing manufacturing facility infrastructure, requiring only the addition of the AI analytics layer demonstrated in this study.

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Project: Predictive Maintenance & Anomaly Detection in 6G Smart Manufacturing — Thales Group